

SARITA SAPKOTA

CIVIL ENGINEERING STUDENT

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PROFILE SUMMARY

A recent Civil Engineering graduate from Kathmandu University passionate about structural engineering and technology-driven structural health monitoring. Experienced in hydropower design, hydraulic modeling, and geospatial analysis, with hands-on fieldwork and deep learning research for crack detection. Committed to advancing resilient and sustainable infrastructure solutions.

RESEARCH EXPERIENCE

Multi Scale Deep Learning for Hairline Crack Detection and Analysis using Hybrid CNN and Transform architecture **Apr 2025 – Jul 2025**
(Submitted for Publication)

- Conducted research under Prof. Dr. Sudan Jha in collaboration with Ravi Kumar Pajiyar, Department of Computer Engineering, Kathmandu University, focusing on deep learning for structural health monitoring.
- Developed a hybrid CNN-Transformer model to accurately detect and classify hairline cracks in concrete, aiming to support predictive maintenance through multiscale feature extraction and global context understanding.

INDUSTRIAL EXPERIENCE

Internship Trainee, TODA Corporation – Japan-Oriented Construction Training Program **Jun 2025 - Aug 2025**

- Participated in a specialized training program for engineers and supervisors conducted by TODA Corporation.
- Engaged in live-site construction work, drew and analyzed drawings, supervised the site, and learned Japanese construction practices.

Internship Trainee, Kabeli A Hydropower Project **Feb 2025 - Apr 2025**

- Worked on a 37.6 MW run-of-river (ROR) hydropower project utilizing the Kabeli River.
 - Assisted in quality checks, reinforcement work, and concrete testing.
 - Supported drawing analysis, prepared BOQ, and report preparation.
- Github Link: [Kabeli-A Internship](#)

CLUB EXPERIENCE

Media Head, Kathmandu University Youth Red Cross Circle **Nov 2024 - Apr 2025**

- Led social media management, content creation, and engagement campaigns; designed promotional materials and documented events.
- Supported website development and enhanced overall digital presence.
- Website link: <https://yrcc.ku.edu.np/>

Executive Member, Indoor Club, Kathmandu University **Mar 2024 - oct 2025**

- Organized skill-based workshops and coordinated inter-departmental tournaments promoting indoor games such as table tennis, badminton, and chess.
- Facilitated leadership initiatives and event management, enhancing student engagement, teamwork, and a healthy campus lifestyle.

General Member, Kathmandu University Civil Engineering Club

Mar 2021 - Mar 2024

- Organized talk shows, treasure hunts, and table tennis tournaments; actively volunteered in various club events.
- Attended technical workshops on municipal drawing and water supply systems to enhance practical skills.
- Website link: <https://cec.ku.edu.np/>

Remote Sanitation and Soil Analysis Volunteer, EWB Nepal , KU Chapter

Dec 2023 - Mar 2024

- Conducted field data collection and infiltration rate testing using a Single Ring Infiltrometer to support septic system planning for a rural school.
- Assessed design flaws and sanitation challenges in existing toilet infrastructure and proposed improvement strategies.
- Website Link: <https://ewb.ku.edu.np/>

PROJECTS

Hydropower Design - Final Year Project

- Conducting hydrological, hydraulic, and structural analysis for Kabeli-A Hydroelectric Project.
- Assessing economic viability and financial aspects for optimized power generation
- Github Link: [Final Year Major Project](#)

RUM Project - Sunkoshi Hydropower

- Performed hydrological analysis and hydraulic design for Sunkoshi Small Hydropower (2.5 MW).
- Optimized water conveyance and sediment management systems.
- Github Link: [Sunkoshi Hydropower](#)

Road Design - Mars , Nepal

- Designed road alignments using SW-Road, ensuring safety, efficiency, and minimal ecological impact.
- Applied sustainable practices to minimize environmental disruption. .
- Github Link: [Mars Nepal Road Design](#)

Demonstrative Model - Types of Footings

- Built a model showcasing different footing types, highlighting structural applications and significance. Enhanced understanding of structural design concepts through hands-on demonstration. Github Link: [Footings-Demonstrative-Model](#)

Trishuli-Hydropower -Topography-Model

- Built a scaled topographic model of Trishuli Hydropower, highlighting key components like dam, penstock, and powerhouse.
- Applied contour mapping, scale conversion, and GIS-based terrain analysis.
- Github Link: [Trishuli-Hydropower-Topography-Model](#)

SKILLS

CAD Tools: AutoCAD 2D and Civil 3D, SW-Road ,Sketchup

Hydraulic & Water Resource Software: EPANET, HEC-RAS

Structural Analysis & Modeling: SAP (Structural integrity and analysis)

Surveying Tools: Total Station, GPS, and other site analysis tools, QGIS

Project Management & Documentation: Microsoft Word, Excel, Project libre, Canva

Version Control: Git and Github

Interests: Structural Engineering, Construction Management, Environmental Engineering & ML

Soft Skills: Effective Communication, Flexible, Fast-Paced, Leadership

EDUCATION

Bachelor of Civil Engineering(Specialization in Hydropower)
Kathmandu University

Feb 2021 - Jun 2025

High Level School in Science

2018 - 2020

Vidhya Mandir Higher Secondary School

Secondary Level School(SEE)

upto 2018

Dhawalagiri Birendra Higher Secondary School

COURSE AND CERTIFICATIONS

Water Supply Training

- Kathmandu University Civil Engineering Club

Going Places with Spatial Analysis

- Esri

Municipal Drawing

- Kathmandu University Civil Engineering Club

AutoCAD 2D & 3D

- SourceCAD

Introduction to Remote Sensing

- Geo University

CONFERENCE AND WORKSHOP ATTENDED

- Attended the Technical Talk Show of Sankalpa '25, with Mr. Kulman Ghising, Former MD of Nepal Electricity Authority, as the Chief Guest, organized by the Department of Civil Engineering, Kathmandu University.
- Attended a talk on Road Development in Nepal by Er. Arjun Jung Thapa, Former Secretary, Ministry of Physical Infrastructure and Transport.
- Attended KUCEC/KUARC Seminar on Seismic-Resistant Construction Practices featuring expert civil engineer Ananta Ram Baidya.
- Attended Basic AutoCAD Workshop]on 4th Semester students organized by KUCEC.
- Attended SANKALPA'25 intro session on "Empowering Future Engineers" focused on open-source GIS tools for water management.
- Attended the OpenGIS Software Tools and Technologies – BEGO program to learn about Giswater for water cycle management.

LEADERSHIP AND VOLUNTEERING EXPERIENCE

- Organized a successful blood donation program during Sankalpa 25 as a member of KUYRCC.
- Volunteered in IT Meet Blood Donation Campaign representing KUYRCC.
- Contributed as a volunteer during the Indreshwari Jatra, supporting event coordination representing KUYRCC
- Contributed to environmental conservation efforts through awareness programs and community engagement in Project Sanrakhxan.
- Organized tree planting and community cleanup drives to promote sustainability During Sankalpa as the annual event of civil engineering (KUCEC).

AWARDS AND HONORS

Hult Prize Participation 2024/2025 We as a group member of 3 participated in the Hult Prize on Campus with our product. Health

Lens that aims to suggest people wisely aligning with SDG 3 (Good Health and Well Being).

Runner Up - Paper Bridge Competition 2023/2024

We, a team of three, participated in the All-Nepal Paper Bridge Competition organized by the Kathmandu University Civil Engineering Club and secured the Runner-Up position.

Outstanding Academic Achievement Award, Sajhedari Shareholder Cooperative Organization Limited, 2018

Recognized for outstanding academic excellence in the S.E.E. Examination, achieving a GPA of 3.35. Award presented for dedication and consistent performance among girl students in the community.

Member at American Society of Civil Engineer (ASCE)

Active ASCE member dedicated to advancing civil engineering through innovation, education, and collaboration. Committed to sustainable infrastructure and public welfare.

REFERENCES

Shyam Sundar khadka - Associate Professor And Associate Dean | Department of Civil Engineering | Kathmandu University | +9779841348522 | sskhadka@ku.edu.np

Sumit Thapa - Project Manager | Kabeli -A Hydropower | +9779843381300 | Mrthapa96@gmail.com